

Optional Office Hours Part III Notes

III.0 Errata

Note that these lecture notes are based off the book called *A Friendly Introduction to Mathematical Logic* by Christopher C. Leary (linked in a future draft) with exposition in these Theorems found in Ch. 4-5. This book contains more detail on the framework (that I'm skipping in these notes) needed to prove the Incompleteness Theorem in full detail.

Additionally, you can check out Enderton's exposition on the Incompleteness Theorem. (in a future draft, I'll provide the necessary sections)

And finally, both a course on Computability Theory/Recursion Theory or a graduate course on mathematical logic usually covers at least one of both incompleteness theorems. (I'll provide an example of some course graduate course material in future notes)

There's also an online resource that I like called *Godel's Theorems without (too many) Tears* that I recommend (and will link in a future draft)

III.1 Godel Numbering and Encoding the Proof System

%EDIT THROUGH THESE NOTES, EDIT THROUGH THE OTHER NOTES, THEN FINISH

For this section and all others, let $\mathcal{L}_{NT} := (0, S, +, \cdot, E, <)$ be defined as the language of number theory

Definition III.1.1. We say that a theory T in the language $\mathcal{L}_{NT} := (0, S, +, \cdot, E, <)$ **satisfies the properties of Basic Arithmetic** if T proves each of the following sentences:

1. $\forall x (\neg Sx) = 0$ (zero not being a successor)
2. $\forall x \forall y (Sx = Sy \rightarrow x = y)$ (injectivity of S)
3. $\forall x (x + 0 = x)$ (additive identity property of 0)
4. $\forall x \forall y (x + Sy = S(x + y))$ (recursive definition of addition)
5. $\forall x (x \cdot 0 = 0)$ (multiplicative property of 0)
6. $\forall x \forall y (x \cdot Sy = x \cdot y + x)$ (recursive definition of multiplication)
7. $\forall x (xE0 = S0)$ (exponential property of 0)
8. $\forall x \forall y (xE(Sy) = (xEy) \cdot x)$ (recursive property of exponentiation)
9. $\forall x (\neg x < 0)$ (least element property of 0)
10. $\forall x \forall y (x < Sy \leftrightarrow (x < y \vee x = y))$ (successor property of inequalities)

11. $\forall x \forall y ((x < y) \vee (x = y) \vee (y < x))$ (*trichotomy law of inequalities*)

We shall call the set of sentences containing precisely axioms 1-11 N .

Definition III.1.2. We state that a set $A \subset \mathbb{N}^k$ (for any $k \geq 0$) is said to be **representable by Basic Arithmetic** (i.e. the theory of N) if there is an $\mathcal{L}_{NT}$ -formula $\phi(x_1, \dots, x_k)$ such that

$$(a_1, \dots, a_k) \in A \iff N \vdash \phi(\overline{a_1}, \overline{a_2}, \dots, \overline{a_k}),$$

where for any $a \in \mathbb{N}$, we define $\overline{a} := \underbrace{S \cdots S}_a 0$.

Recall in **Section I.2** of the [Optional Office Hours Part I Notes](#) that we define a subset of $\mathbb{N}^k$ that is decideable if there exists an "effective procedure", aka, an algorithm that we can execute as a computer program that determines whether or not something is in that set.

What we've stated above is an equivalent condition of a decideable set that is useful for the purposes of proving the incompleteness theorem.

Theorem III.1.3. A set $A \subset \mathbb{N}^k$ (for any $k \geq 0$) is decideable if and only if it is representable by Basic Arithmetic. Additionally, we say that a function $f : \mathbb{N}^k \rightarrow \mathbb{N}$ is representable if its graph $\text{graph}(f) \subset \mathbb{N}^{k+1}$ is representable by Peanno Arithmetic, and hence a function f (i.e. $\text{graph}(f)$) is decideable if and only if $\text{graph}(f)$ is representable by Peanno Arithmetic.

Proof. Outside the scope of the course, since advanced concepts in computability/recursion theory are needed to rigorously prove this result. $\square$

From this point forward, we shall use "decideable set/function" (**Definition I.2.1.** of the **Section I.2** of the [Optional Office Hours Part I Notes](#)) and a "representable set/function by Peanno Airthmetic" (**Definition III.1.2**) interchangeably.

So now we'll mention a few results on what is decideable, which is proven through showing existence of a code in $\mathcal{L}_{NT}$ where we can enumerate each object in a way we can then talk about in $\mathcal{L}_{NT}$

Note that we can code finite sequences $S := \langle k_1, \dots, k_n \rangle$ in $\mathcal{L}_{NT}$ using prime factorization as so:

$\alpha_S := 2^{k_1} \cdot 3^{k_2} \cdot \dots \cdot p_n^{k_n}$, where n is the n th least prime number.

Lemma III.1.4. The collection of numbers that are codes for finite sequences are a decideable set.

Outline of the proof. Note that we are able to talk about prime factorizations using $\mathcal{L}_{NT}$, which shows representability by Peanno arithmetic, showing decideability. $\square$

Now using this notion of sequences, we can elaborate our code to talk about $\mathcal{L}_{NT}$ -formulas using (in post-script notation for atomic formulas) finite sequences of numbers and coding a number for every symbol, i.e., something like:

Symbol Symbol Number

$\neg$	1
$\rightarrow$	3
$\forall$	5
$=$	7
0	9
S	11
$+$	13
$\cdot$	15
E	17
$<$	19
$($	21
$)$	23
v_i	$2i$

So for example $= 0S0$ would be coded as the sequence

$\langle 7, 9, 11, 9 \rangle$,

which would be coded as the number

$2^7 3^9 5^{11} 7^9$,

Which according to Christopher O'Leary (take that silly author's word for it!) would explicitly be the number

4964250291131250000000.

Using this algorithm of coding a $\mathcal{L}_{NT}$ -formula as a number, this yields the following definition of the Godel number.

Definition III.1.5. Given a formula ϕ , we define the **Godel Number** $\ulcorner \phi \urcorner$ of a $\mathcal{L}_{NT}$ -formula ϕ to be the number defined by the function $\ulcorner - \urcorner : \mathcal{L}_{NT}\text{-strings} \rightarrow \mathbb{N}$ recursively as follows:

$$\ulcorner s \urcorner := \begin{cases} 2^{13} 3^{\ulcorner \alpha \urcorner} & \text{if } s \text{ is } (\neg \alpha) \text{ where } \alpha \text{ is an } \mathcal{L}_{NT}\text{-formula} \\ 2^3 3^{\ulcorner \alpha \urcorner} 5^{\ulcorner \beta \urcorner} & \text{if } s \text{ is } (\alpha \rightarrow \beta) \text{ where } \alpha, \beta \text{ are } \mathcal{L}_{NT}\text{-formula} \\ 2^5 3^{\ulcorner v_i \urcorner} 5^{\ulcorner \alpha \urcorner} & \text{if } s \text{ is } (\forall v_i \alpha) \text{ where } \alpha \text{ is an } \mathcal{L}_{NT}\text{-formula} \\ 2^7 3^{\ulcorner t_1 \urcorner} 5^{\ulcorner t_2 \urcorner} & \text{if } s \text{ is } = t_1 t_2 \text{ where } t_1, t_2 \text{ are } \mathcal{L}_{NT}\text{-terms} \\ 2^9 & \text{if } s \text{ is } 0 \\ 2^{11} 3^{\ulcorner t \urcorner} & \text{if } s \text{ is } St, \text{ with } t \text{ a term} \\ 2^{13} 3^{\ulcorner t_1 \urcorner} 5^{\ulcorner t_2 \urcorner} & \text{if } s \text{ is } + t_1 t_2, \text{ where } t_1 \text{ and } t_2 \text{ are terms} \\ 2^{15} 3^{\ulcorner t_1 \urcorner} 5^{\ulcorner t_2 \urcorner} & \text{if } s \text{ is } \cdot t_1 t_2, \text{ where } t_1 \text{ and } t_2 \text{ are terms} \\ 2^{17} 3^{\ulcorner t_1 \urcorner} 5^{\ulcorner t_2 \urcorner} & \text{if } s \text{ is } Et_1 t_2, \text{ where } t_1 \text{ and } t_2 \text{ are terms} \\ 2^{19} 3^{\ulcorner t_1 \urcorner} 5^{\ulcorner t_2 \urcorner} & \text{if } s \text{ is } < t_1 t_2, \text{ where } t_1 \text{ and } t_2 \text{ are terms} \\ 2^{2i} & \text{if } s \text{ is the variable } v_i \\ 3 & \text{otherwise.} \end{cases}$$

Lemma III.1.6.

- (i) A function $f : \mathbb{N}^k \rightarrow \mathbb{N}$ defined recursively by decideable functions is decideable.
- (ii) The Godel Numbers of $\mathcal{L}_{NT}$ -formulas are decideable.

Outline of proof. **Part (i)** follows by applying the recursive definition of the base function alongside the utilization of recursion which is well-known to be something that can be done in computing. **Part (ii)** More or less follows by the recursive algorithm coding for a Godel Number done recursively in **Definition III.1.5**, and **Part (i)** allowing us to do so. $\square$

Next, we can code deductions using sequences as so:

Definition III.1.7. Given a deduction $D := \langle \phi_1, \phi_2, \dots, \phi_k \rangle$, we define the **sequence code** of a deduction D to be the number

$$\ulcorner D \urcorner := 5^{\ulcorner \phi_1 \urcorner} 7^{\ulcorner \phi_2 \urcorner} \dots p_{k+2}^{\ulcorner \phi_k \urcorner},$$

where p_{k+2} is the $k + 2$ th least prime number.

Lemma III.1.8. The set of sequence codes of deductions of a given decidable theory T are decidable.

Outline of proof. Immediate from previous results and using prime number factorizations, along with the ability to verify whether the formula of a sequence is a logical/nonlogical axioms (using decidability of Δ and T as hypothesized) or follows from Modens Ponens of previous entries. $\square$

Corollary III.1.9. Whether or not a sequence of formulas (encoded in $\mathbb{N}$) is a correct proof is decidable.

Proof. Restatement of **Lemma III.1.8.** $\square$

To recap everything in this section, we have shown that the language of $\mathcal{L}_{NT}$ can talk about the $\mathcal{L}_{NT}$ -terms, formulas, and proofs, themselves are decidable, and moreover we showed that one is able to code a proof verification system as to whether a "proof" is correct or not.

This shows something, which is itself very significant, which is that mathematics/computing can itself verify whether a given "proof" is correct. There is a catch to this, which is that the math itself must be encoded into some computer system, and all of mathematics is discussed using codes for these proof systems.

The problem with executing this theory into practice is a few things:

1. We need to make sure our programs can be done in some reasonable amount of time (there's no point if a given program takes 1,000,000 years to execute!)
2. We need mathematicians to universally agree on a given computer-coded language to write math (good luck getting a whole field of academia to universally agree to collectively do anything!)

So we as mathematicians are at the whims of peer reviewers, and our ability to automate all of mathematics (in particular the peer review system) using some kind of "AI algorithm" currently falls flat.

III.2 Incompleteness and Undecidability Results

The previous section indicates the power of mathematics to do a lot of things, so even though we laugh at it now, it isn't particularly absurd in hindsight that a significant portion of early 20th century mathematicians (most Notably David Hilbert) thought that Peanno Airthmetic, or any theory that proves Peanno Arithmetic, is decideable and complete (and possibly with results like the set of formulas true in the set natural numbers $\mathbb{N}$ being decideable)

It turns out, as I'm sure you guessed, that this result is a negative. To outline the proof, we need one more result, the grand finale of all the lemmas I give in these notes prior to the incompleteness theorem: The Self-Reference Lemma

Lemma III.2.1. (Godel's Self-Reference Lemma) Let $\psi(x)$ be a $\mathcal{L}_{NT}$ -formula with only x free. Then there is a sentence ϕ such that

$$N \vdash (\phi \leftrightarrow \psi(\overline{\ulcorner \phi \urcorner})).$$

Proof. WAAAYYY OUTSIDE THE SCOPE OF THESE NOTES. Refer to *page 182 of Friendly Introduction to Mathematical Logic by Christopher C Leary* for an actual proof. $\square$

What this important lemma tells us is we can prove a formula that is logically equivalent to a property that we can give about itself, allowing us to talk about formulas that have self-referencing characteristics.

Theorem III.2.2. (Godel's First-Incompleteness Theorem) Let T be a consistent decideable theory that proves Basic Arithmetic, i.e., we have $T \vdash \phi$, for all $\phi \in N$. Then T is incomplete, i.e., there is a $\mathcal{L}_{NT}$ -formula θ that is true in $\mathfrak{N} := (\mathbb{N}, S, +, \cdot, E, <)$ such that $T \not\vdash \theta$ and $T \not\vdash \neg\theta$.

Outline of proof. Using **Lemma III.2.8**, let $\text{Deduction}_T(x, y)$ be the fomula stating that x is the code for a proof of a formula y , i.e., it states that $x = \ulcorner D \urcorner$ and $y = \ulcorner \phi \urcorner$, such that D is a deduction using T to an $\mathcal{L}_{NT}$ -formula ϕ . We then define

$$\text{Thm}_T(x) := \exists y \left(\text{Deduction}_T(y, x) \right),$$

which we can verify (details left to the reader) states that T proves a given Godel Number of a formula x , i.e., for every $\mathcal{L}_{NT}$ -formula ϕ , we have

$$T \vdash \phi \iff \mathfrak{N} \models \text{Thm}_T(\overline{\ulcorner \phi \urcorner}). \quad (III.1)$$

So using **Lemma III.2.1 (The Self-Reference Lemma)**, we find there exists some $\mathcal{L}_{NT}$ -formula θ such that

$$T \vdash \theta \leftrightarrow \neg \text{Thm}_T(\ulcorner \theta \urcorner). \quad (III.2)$$

We can then conclude (using (III.1) with additional steps left to the reader) that if (towards contradiction) T is complete, then using consistency, we have

$$T \vdash \theta \iff T \nvdash \theta,$$

which is obviously a contradiction in either case where $T \vdash \theta$ or $T \nvdash \theta$. $\square$

Note that this proof utilizes an idea in philosophy called the "Liar paradox" which is the paradox of truth where we make a statement self-referencing the statement itself and stating that it's a lie, i.e.

"This statement is a lie",

resulting in a statement that implies its falsehood if true and vice versa, giving us a paradox. Godel's proof cleverly used the self-reference lemma to formulate the liar paradox using self-reference formulate such a paradox-contradiction if towards contradiction T was complete, allowing us to do formulate a counterexample of something that can't be provable, making a decideable theory incomplete.

Speaking of the liar paradox, this self-reference lemma strategy allows us to not just prove the incompleteness of undecidable theories but also the undefinability of truth in $\mathfrak{N}$, since the definability of truth would allow us to use the self-reference lemma to formulate the liar paradox.

Theorem III.2.3. (Tarski's Theorem aka the undefinability of truth) The set of Godel formulas true in $\mathfrak{N}$ are undefinable in $\mathfrak{N}$.

Proof. Suppose towards contradiction there exists a formula $\theta(x)$ (with only x has free) such that

$$\mathfrak{N} \models \theta(\ulcorner \phi \urcorner) \iff \mathfrak{N} \models \phi.$$

I will leave it to the reader to use the self-reference lemma to construct a formula ξ such that

$$\mathfrak{N} \models \xi \iff \mathfrak{N} \not\models \xi,$$

and verify the details, giving us a contradiction. $\square$

Definition III.2.4. The **Axioms of Peanno Arithmetic**, or PA , are all eleven axioms of N (see **Definition III.1.1**) together with the axiom scheme

$(\phi(0) \wedge \forall x(\phi(x) \rightarrow \phi(Sx))) \rightarrow \forall x\phi(x)$, for each $\mathcal{L}_{NT}$ -formula $\phi(x)$ with a single free variable.

showing that every $\mathcal{L}_{NT}$ -formula with a free variable holds universally if and only if it satisfies the "principle of mathematical induction" formalized in the language of number theory.

Using such axioms of Peanno-Arithmetic, we can derive the following results

Lemma III.2.5. Let ϕ be a $\mathcal{L}_{NT}$ -formula. Then

(i) If $PA \vdash \phi$, then $PA \vdash \text{Thm}_{PA}(\overline{\ulcorner \phi \urcorner})$

(ii) $PA \vdash \text{Thm}_{PA}(\overline{\ulcorner \phi \urcorner}) \rightarrow \text{Thm}_{PA}(\overline{\text{Thm}_{PA}(\overline{\ulcorner \phi \urcorner})})$

(iii) $PA \vdash \text{Thm}_{PA}(\overline{\ulcorner \phi \urcorner}) \wedge \text{Thm}_{PA}(\overline{\ulcorner \phi \rightarrow \psi \urcorner}) \rightarrow \text{Thm}_{PA}(\overline{\ulcorner \psi \urcorner})$

%GIVE AN OUTLINE OF THIS PROOF

Given a theory T , define Con_T as the sentence $\neg \text{Thm}_T(\overline{\ulcorner \perp \urcorner})$, where $\perp := \forall v_1(\neg v_1 = v_1)$, which essentially tells us a theory is consistent. I'll leave it to the reader using **Lemma III.2.5** to show that

$$T \vdash \text{Con}_T \iff T \text{ is consistent.}$$

%GIVE MORE ELABORATE OUTLINE OF THIS PROOF

Theorem III.2.6. (*Godel's Second Incompleteness Theorem*) If T is a consistent decidable theory such that T is consistent, then $T \not\vdash \text{Con}_T$.

Outline of Proof. I'll give a more detailed proof in an update of these notes, but for now I'll give the general strategy, which is to show that

$$T \vdash \text{Con}_T \leftrightarrow \theta,$$

where θ is the formula determined by applying the **Self-Reference Lemma** to (III.2), which leads to a contradiction since we proved in the **First Incompleteness Theorem (Theorem III.2.2)** that $T \not\vdash \theta$, hence we have $T \not\vdash \text{Con}_T$. $\square$

Note that this result has even more general ramifications for higher mathematics, in particular most of modern mathematics that utilizes set theory.

Corollary III.2.7. If the ZFC axioms of set theory (discussed in **Section II.3** of the [Optional Office Hours Part II Notes](#)) are consistent, then we cannot prove the consistency of ZFC.

Outline of Proof. Note that using the ordinal number construction given in (II.2) of **Section II.2** of the Optional Office Hours Part II Notes, we can encode the natural numbers in the language of set theory, and it can be shown that these encoded natural numbers have all the properties of Peano Arithmetic, and hence our conclusion follows immediately by **Godel's Second Incompleteness Theorem**. $\square$

What this result tells us is not only that we have no hope of coming up with a decidable axiom scheme of arithmetic where we can prove its own consistency, but neither can any of higher mathematics that can talk about Peano Arithmetic (such as Set Theory and basically any other mathematical language that we formulate that meaningfully talks about mathematics).

%Peano Arithmetic and the Second incompleteness theorem
Theorem 5.6.3

%make corollary about not being able to show set theory is consistent if it is consistent